CLASS 9 — SCIENCE · FORMULA SHEET

Chapter 2: **Exploring Mixtures and Their Separation** | Quick revision, one page

Matter and mixtures

- Pure substance: single kind of particle, fixed composition and fixed melting/boiling point (elements and compounds).
- Mixture: two or more substances mixed in any proportion; components keep their properties.
- Homogeneous mixture (solution, e.g. brass, air): uniform composition. Heterogeneous mixture (e.g. sand and iron filings): visibly different parts.

Solutions, colloids, suspensions

- Solution: solute particles < 1 nm, do not scatter light, do not settle, pass through filter paper.
- Colloid (e.g. milk, fog): particles 1 to 1000 nm, scatter light (Tyndall effect), do not settle, not separated by filtration.
- Suspension (e.g. muddy water): particles > 1000 nm, scatter or block light, settle on standing, separated by filtration.

Concentration

- Mass per cent $= \dfrac{\text{mass of solute}}{\text{mass of solution}} \times 100$, where mass of solution = mass of solute + mass of solvent.
- Volume per cent $= \dfrac{\text{volume of solute}}{\text{volume of solution}} \times 100$.
- Saturated solution: cannot dissolve more solute at that temperature. Solubility of most solids increases with temperature; of gases decreases.

Separation methods and principle

- Filtration: insoluble solid from liquid. Evaporation: dissolved solid from solvent. Crystallisation: pure crystals of a dissolved solid.
- Simple distillation: liquid from a non-volatile solute. Fractional distillation: miscible liquids with boiling points differing by < 25 K.
- Separating funnel: immiscible liquids. Centrifugation: denser particles settle (cream from milk). Chromatography: differing solubilities (dyes in ink).
- Sublimation: e.g. ammonium chloride, camphor from a non-subliming solid. Magnetic separation: iron from sulphur or sand.

Dilution

- Mass of solute is unchanged on dilution: $\text{per cent}_1 \times \text{mass}_1 = \text{per cent}_2 \times \text{mass}_2$.